

Group Coursework Submission Form

Specialist Masters Programme

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COURSEWORK 2

City St George's, University of London
BAYES BUSINESS SCHOOL

Financial Economics (CM2)

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Question 1

(i) Using 10 years of historical daily stock price data for Microsoft Corporation provided in the file "*CM2_Coursework2_Group10.xlsx*", the following statistics have been calculated in Excel, using functions for sample variance and standard deviation as well as the **annualized standard deviation** given by the formula.

standard deviation (annualized) = standard deviation of daily logarithmic returns * $\sqrt{252}$

- Variance log-returns (daily) = 0.000289907
- StDev log-returns (daily) = 0.017026669
- StDev (annualized) = σ = 0.270289989

(ii) Let denote by:

- $T = 1$ is the total time horizon of the model (in years) for a 1-year tree
- $N = 10$ is the number of the time steps
- $dt = \Delta t$ is length of each time step (in years)

The time step used is calculated using the formula:

$$\Delta t = \frac{T}{N} = \frac{1}{10} = 0.1 \text{ years} \quad (1)$$

To model the evolution of Microsoft Corporation's stock price over time, a binomial tree is constructed and calibrated using the Cox-Ross-Rubinstein (CRR) model.

The CRR framework assumes that at each time step, the stock price can increase by a factor of u or decrease by d , with the corresponding probabilities. These **up- and down-factors** are derived from the formulas.

$$u = e^{\sigma\sqrt{\Delta t}} = 1.089232369$$
$$d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}} = 0.918077748$$

where:

- σ is the annualized volatility of the stock
- Δt is the length of each time step in years

The **risk-neutral probability** q of an upward movement is then computed by:

$$q = \frac{e^{r\Delta t} - d}{u - d} = 0.500889 \quad (2)$$

Question 2

The CRR binomial model was implemented in R to simulate the evolution of Microsoft's stock price over 10 time steps (1 year in total). The initial stock price is set to \$408.21, based on the most recent value from the historical data set provided. The model parameters also include an annualized volatility σ of approximately 27%.

The up and down multipliers (u and d) are derived from the volatility and time step size in R as follows:

```
1 S0 <- 408.21          # Initial stock price
2 N <- 10               # Number of time steps
3 T <- 1                # Total time (in years)
4 dt <- T / N           # Time step
5 sigma <- 0.270289989  # Annualized volatility
6
7 # CRR up and down factors
8 u <- exp(sigma * sqrt(dt))
9 d <- 1 / u
```

The code below builds the binomial tree by iteratively calculating possible stock prices at each node, based on the number of up and down moves:

```
1 # Build the stock price tree
2 nodes <- data.frame()
3
4 for (i in 0:N) { #i: time step
5   for (j in 0:i) { #j:how many times the price went down
6     price <- S0 * u^(i - j) * d^j #i-j:how many times it went
7     up
8     nodes <- rbind(nodes, data.frame(time = i, level = j, price
9     = price)) }}
10
11 # Create the connections between nodes
12 edges <- data.frame()
13
14 for (i in 0:(N - 1)) {
15   for (j in 0:i) {
16     current <- nodes[nodes$time == i & nodes$level == j, ]
17
18     # Up move
19     up <- nodes[nodes$time == i + 1 & nodes$level == j, ]
20     edges <- rbind(edges, data.frame(
21     x = current$time, y = current$price,
22     xend = up$time, yend = up$price,
23     color = "Up" ))
24
25     # Down move
26     down <- nodes[nodes$time == i + 1 & nodes$level == j + 1, ]
27     edges <- rbind(edges, data.frame(
28     x = current$time, y = current$price,
29     xend = down$time, yend = down$price,
30     color = "Down" )) } }
```

The resulting tree is plotted using the package ggplot2, with upward moves shown in blue and downward moves in red:

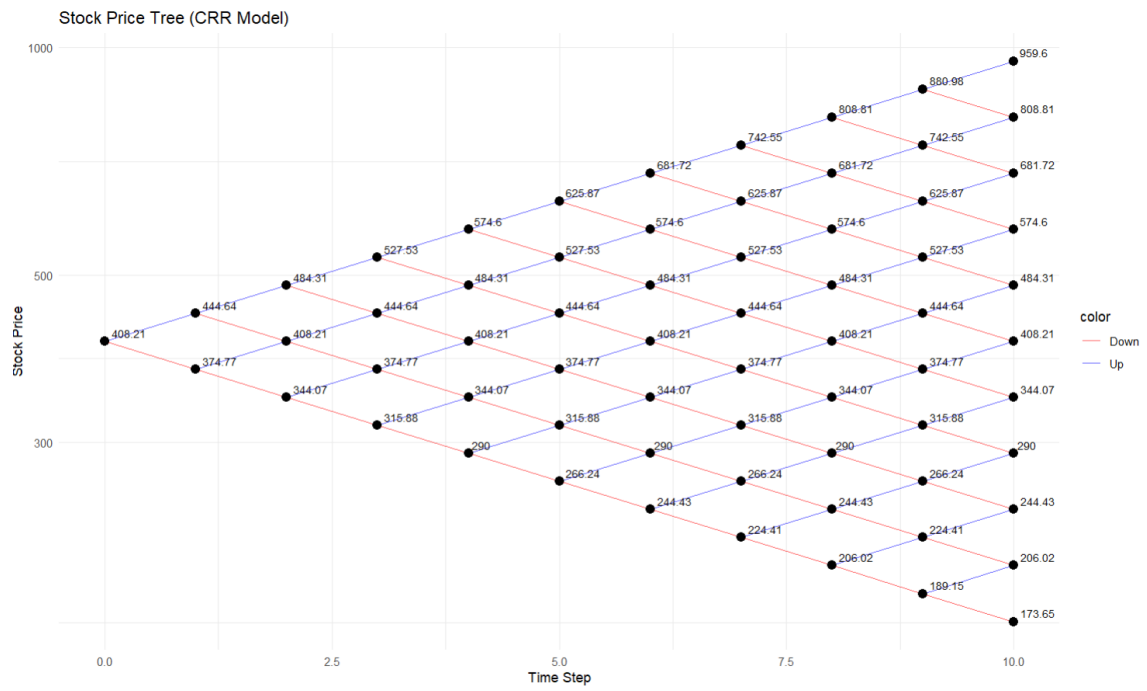


Figure 1: Binomial tree of Stock price

Question 3

A **European plain vanilla put option** with a maturity of one year and a strike price equal to 105% of the initial stock price ($105\% \cdot 408.21$) is evaluated using the CRR binomial tree model calibrated in Q2 using R.

The calculations start off with a function that constructs the stock price tree, where each node represents a possible future price based on the number of up (u) and down (d) moves over N time steps:

```
1 make_tree <- function(S0, u, d, N) {  
2   tree <- matrix(0, N+1, N+1)  
3   for (i in 0:N) for (j in 0:i) tree[j+1,i+1] <- S0 * u^j * d^(  
4     i-j)  
5   tree }  
6 }
```

At maturity T (final column of the tree), the payoff of the European put is calculated as

$$\max(K - S_T, 0) \quad (3)$$

and stored in the *euro_tree*:

```
1 price_tree <- make_tree(S0, u, d, N)  
2 euro_tree <- matrix(0, N+1, N+1)  
3 for (j in 0:N) {  
4   payoff <- max(K - price_tree[j+1,N+1], 0)  
5   euro_tree[j+1,N+1] <- payoff  
6 }
```

The next loop performs backward induction: for each earlier time step, the option's value is calculated as the discounted expected value under the risk-neutral measure using probability q

```
1 for (i in (N-1):0) for (j in 0:i) {  
2   cont <- exp(-r*dt)*(q*euro_tree[j+2,i+2] + (1-q)*euro_tree[j  
3     +1,i+2])  
4   euro_tree[j+1,i+1] <- cont  
5 }
```

If the option price is V , we go backwards through the tree to calculate the present value of the option at each node using the formula

$$V_{i,j} = e^{-r \cdot \Delta t} [q \cdot V_{i+1,j+1} + (1 - q) \cdot V_{i+1,j}] \quad (4)$$

The value at the root of the tree (*euro_tree*[1,1]) gives the present value of the European put option

```
1 cat("European Put Price at t=0:", round(euro_tree[1,1], 4), "\n"  
2   )  
3 European Put Price at t=0: 47.0053
```

The tree is visualized using ggplot2, with each node labeled by the computed option value, helping to illustrate the pricing dynamics over time.

```
1 print(plot_tree(euro_tree, N, "European"))
```

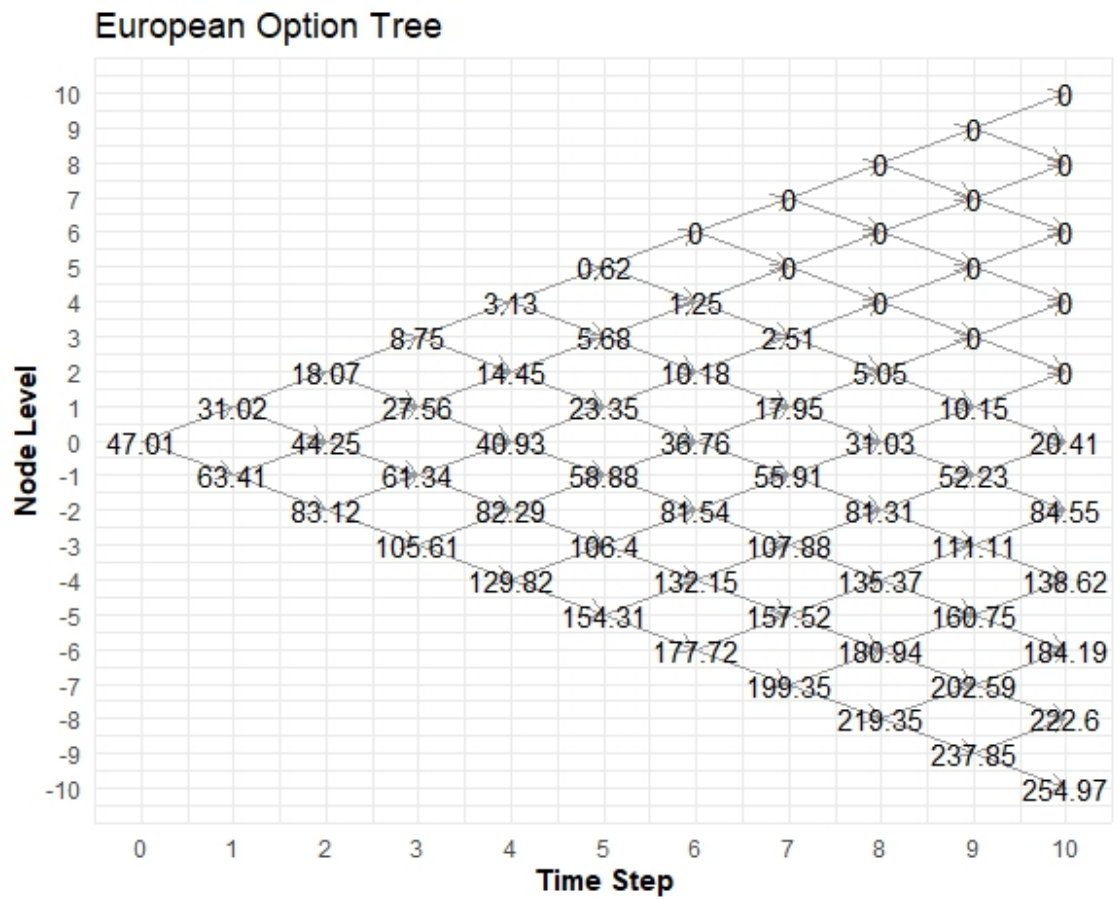


Figure 2: European option tree

Question 4

Building on the **European put option** from Question 3, we now consider a **Bermudan put option** with identical parameters, except that it allows early exercise at **10 discrete time steps**. This introduces a crucial difference in the valuation process: at each node in the binomial tree, we compare the **continuation value** (the expected discounted value of holding the option) with the **early exercise value** (the immediate payoff), and take the maximum.

Mathematically, for each node (i, j) , the Bermudan option value is given by:

$$V_{i,j} = \max \left(e^{-r\Delta t} [q \cdot V_{i+1,j+1} + (1-q) \cdot V_{i+1,j}], K - S_{i,j} \right) \quad (5)$$

The early exercise value $K - S_{i,j}$ represents the immediate payoff if the option is exercised at that node, meaning the holder receives the strike price K and sells the asset at $S_{i,j}$, provided this results in a positive payoff. This is implemented in R by modifying the backward induction step as follows:

```
1 for (i in (N-1):0) for (j in 0:i) {
2   cont2 <- exp(-r*dt)*(q*berm_tree[j+2,i+2]+(1-q)*berm_tree[j
3     +1,i+2]) # Continuation value
4   early <- max(K - price_tree[j+1,i+1], 0)
5     # Early exercise value
6   berm_tree[j+1,i+1] <- max(cont2, early)
7     # Option value at node
8 }
```

The value at the root of the tree gives the present value of the Bermudan put option:

```
1 cat("Bermudan Put Price at t=0:", round(berm_tree[1,1], 4), "\n")
2 # Bermudan Put Price at t=0: 49.3266
```

To visualize the evolution of option values, we use `ggplot2`:

```
1 print(plot_tree(berm_tree, N, "Bermudan"))
```

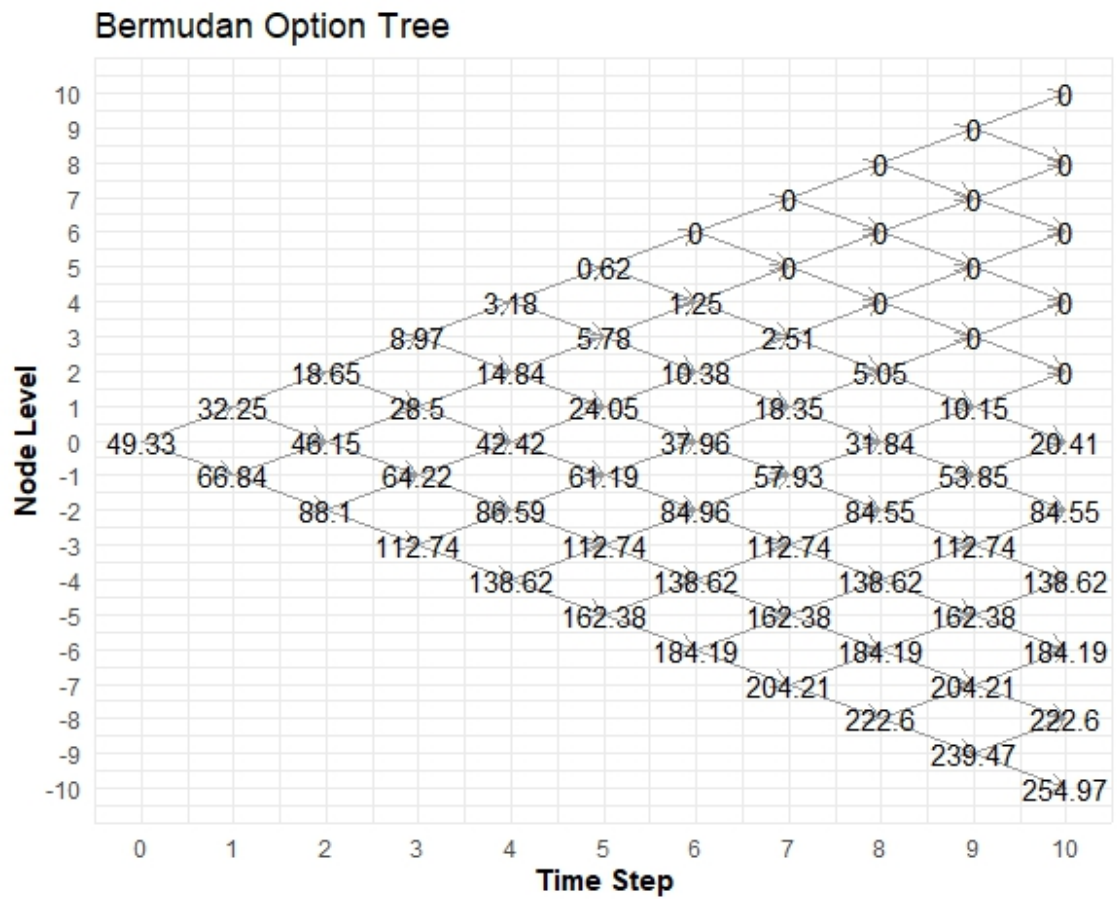


Figure 3: Bermudan option tree

Question 5

The Bermudan put option (\$49.33) is more expensive than the European put option (\$47.01), because it allows traders the right to exercise early at any of the 10 steps when the option is significantly in the money. The \$2.32 premium represents the additional value of this flexibility, particularly in volatile markets (27% in our example). For example, at nodes where stock price drops (e.g time 2, node -2) the Bermudan option has a higher intrinsic value (\$88.10 vs \$83.12) and similarly at nodes where stock price rises (e.g time 2, node 2) the Bermudan option still outperforms, but by less (\$18.65 vs \$18.07). However, when the options are close to being worthless (e.g. time step 5, node 5), the option values are the same since early exercise provides no benefit. At maturity ($t = 10$), the final option prices are also the same as they depend only on the stock price at that time.

Question 6

As the number of exercise times increases, the price of Bermudan options will increase strictly. This is because additional exercise nodes give more opportunities to exercise early when the option is deep in the money (intrinsic value $>$ continuation value). Furthermore, the Bermudan price will approach but never hit the American option price (maximum option price with continuous exercise rights). This is because increasing the time steps will not completely eliminate the opportunity cost of having discrete exercise compared to continuous exercise.